SOLUTIONS TO END-OF-CHAPTER PROBLEMS

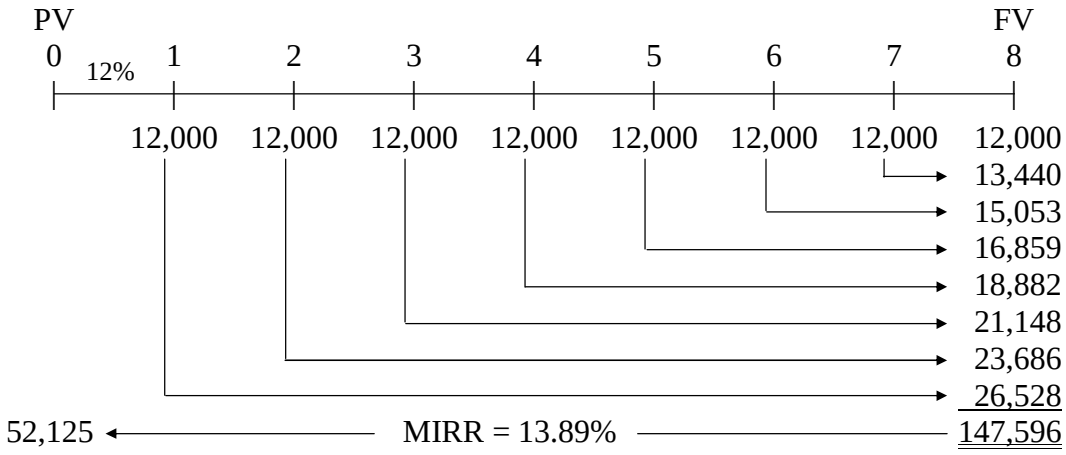
10-1 $NPV = -\$52{,}125 + \$12{,}000[(1/I) - (1/(I \times (1 + I)^N)]$
$= -\$52{,}125 + \$12{,}000[(1/0.12) - (1/(0.12 \times (1 + 0.12)^8)]$
$= \$7{,}486.68.$

Financial calculator solution: Input $CF_0 = -52125$, $CF_{1-8} = 12000$, I/YR = 12, and then solve for NPV = $7,486.68.

10-2 Financial calculator solution: Input $CF_0 = -52125$, $CF_{1-8} = 12000$, and then solve for IRR = 16%.

10-3 MIRR: PV Costs = $52,125.

FV Inflows:



Financial calculator: Obtain the FVA by inputting N = 8, I/YR = 12, PV = 0, PMT = 12000, and then solve for FV = $147,596. The MIRR can be obtained by inputting N = 8, PV = -52125, PMT = 0, FV = 147596, and then solving for I/YR = 13.89%.

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10-4 $PV = \$12{,}000[(1/I) - (1/(I \times (1 + I)^N)]$
$= \$12{,}000[(1/0.12) - (1/(0.12 \times (1 + 0.12)^8)]$
$= \$59{,}611.68.$

Financial calculator: Find present value of future cash flows by inputting N = 8, I/YR = 12, PMT = -12000, FV = 0, then solve for PV = $59,611.68.

PI = PV of future cash flows/Initial cost
= $59,611.68/$52,125 = 1.14.

10-5 Since the cash flows are a constant $12,000, calculate the payback period as: $52,125/$12,000 = 4.3438, so the payback is about 4 years.

10-6 The project's discounted payback period is calculated as follows:

Year	Annual CF	Discounted CF (@12%)	Cumulative Discounted CF
0	-52,125		-52,125.00
1	12,000	10,714.29	(41,410.71)
2	12,000	9,566.33	(31,844.39)
3	12,000	8,541.36	(23,303.02)
4	12,000	7,626.22	(15,676.81)
5	12,000	6,809.12	(8,867.69)
6	12,000	6,079.57	(2,788.11)
7	12,000	5,428.19	2,640.08
8	12,000	4,846.60	7,486.68

The discounted payback period is $6 + \frac{\$2{,}788.11}{\$5{,}428.19}$ years, or 6.51 years.

10-7 a. Project A: Using a financial calculator, enter the following:

$CF_0 = -15000000$
$CF_1 = 5000000$
$CF_2 = 10000000$
$CF_3 = 20000000$

I/YR = 10; NPV = $12,836,213.
Change I/YR = 10 to I/YR = 5; NPV = $16,108,952.
Change I/YR = 5 to I/YR = 15; NPV = $10,059,587.

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Project B: Using a financial calculator, enter the following:

CF_0 = -15000000
CF1 = 20000000
CF2 = 10000000
CF3 = 6000000

I/YR = 10; NPV = $15,954,170.
Change I/YR = 10 to I/YR = 5; NPV = $18,300,939.
Change I/YR = 5 to I/YR = 15; NPV = $13,897,838.

b. Using the data for Project A, enter the cash flows into a financial calculator and solve for IRR_A = 43.97%. The IRR is independent of the WACC, so IRR doesn't change when the WACC changes.

Using the data for Project B, enter the cash flows into a financial calculator and solve for IRR_B = 82.03%. Again, the IRR is independent of the WACC, so IRR doesn't change when the WACC changes.

10-8 *Truck:*

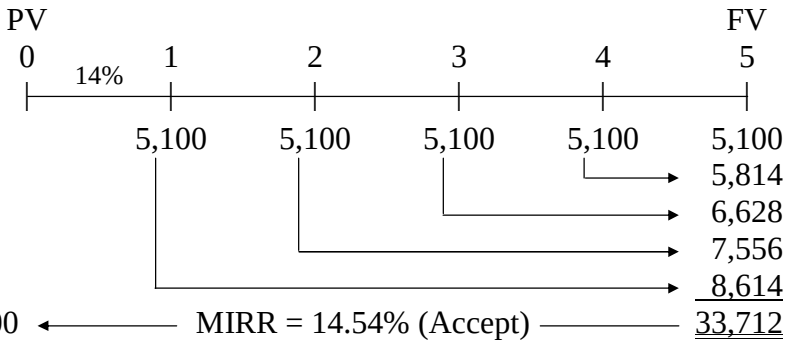
NPV = -$17,100 + $5,100($PVIFA_{14\%,5}$)
= -$17,100 + $5,100(3.4331) = -$17,100 + $17,509
= $409. (Accept)

Financial calculator: Input the appropriate cash flows into the cash flow register, input I/YR = 14, and then solve for NPV = $409.

Financial calculator: Input the appropriate cash flows into the cash flow register and then solve for IRR = 14.99% ≈ 15%.

MIRR: PV Costs = $17,100.

FV Inflows:



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Financial calculator: Obtain the FVA by inputting N = 5, I/YR = 14, PV = 0, PMT = 5100, and then solve for FV = $33,712. The MIRR can be obtained by inputting N = 5, PV = -17100, PMT = 0, FV = 33712, and then solving for I/YR = 14.54%.

Pulley:

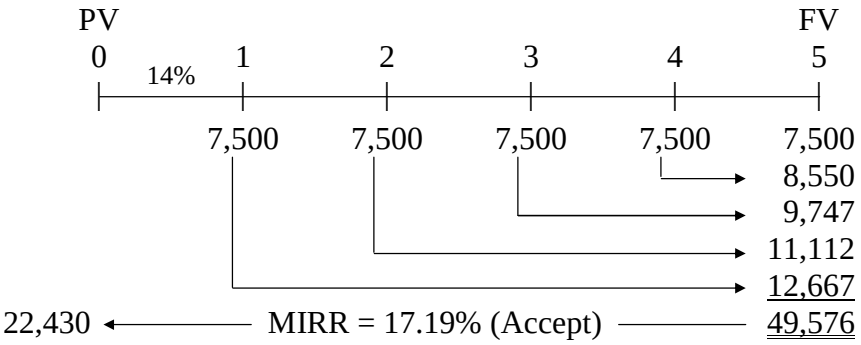
NPV = -$22,430 + $7,500(3.4331) = -$22,430 + $25,748
= $3,318. (Accept)

Financial calculator: Input the appropriate cash flows into the cash flow register, input I/YR = 14, and then solve for NPV = $3,318.

Financial calculator: Input the appropriate cash flows into the cash flow register and then solve for IRR = 20%.

MIRR: PV Costs = $22,430.

FV Inflows:



Financial calculator: Obtain the FVA by inputting N = 5, I/YR = 14, PV = 0, PMT = 7500, and then solve for FV = $49,576. The MIRR can be obtained by inputting N = 5, PV = -22430, PMT = 0, FV = 49576, and then solving for I/YR = 17.19%.

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10-9 *Electric-powered:*

$NPV_E = -\$22,000 + \$6,290[(1/i) - (1/(i \times (1 + i)^n)]$
$= -\$22,000 + \$6,290[(1/0.12) - (1/(0.12 \times (1 + 0.12)^6)]$
$= -\$22,000 + \$6,290(4.1114) = -\$22,000 + \$25,861 = \$3,861.$

Financial calculator: Input the appropriate cash flows into the cash flow register, input I/YR = 12, and then solve for NPV = $3,861.

Financial calculator: Input the appropriate cash flows into the cash flow register and then solve for IRR = 18%.

Gas-powered:

$NPV_G = -\$17,500 + \$5,000[(1/i) - (1/(i \times (1 + i)^n)]$
$= -\$17,500 + \$5,000[(1/0.12) - (1/(0.12 \times (1 + 0.12)^6)]$
$= -\$17,500 + \$5,000(4.1114) = -\$17,500 + \$20,557 = \$3,057.$

Financial calculator: Input the appropriate cash flows into the cash flow register, input I/YR = 12, and then solve for NPV = $3,057.

Financial calculator: Input the appropriate cash flows into the cash flow register and then solve for IRR = 17.97% ≈ 18%.

The firm should purchase the electric-powered forklift because it has a higher NPV than the gas-powered forklift. The company gets a high rate of return (18% > r = 12%) on a larger investment.

10-10 Financial calculator solution, NPV:

Project S

Inputs	5	12		3000	0
	N	I/YR	PV	PMT	FV
Output			= -10,814.33		

$NPV_S = \$10,814.33 - \$10,000 = \$814.33.$

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Project L

Inputs	5	12		7400	0
	N	I/YR	PV	PMT	FV
Output			= -26,675.34		

NPV_L = \$26,675.34 – \$25,000 = \$1,675.34.

Financial calculator solution, IRR:
Input CF_0 = -10000, CF_1 = 3000, N_j = 5, IRR_S = ? IRR_S = 15.24%.
Input CF_0 = -25000, CF_1 = 7400, N_j = 5, IRR_L = ? IRR_L = 14.67%.

Financial calculator solution, MIRR:
Project S

Inputs	5	12	0	3000	
	N	I/YR	PV	PMT	FV
Output					= -19,058.54

$PV\ costs_S$ = \$10,000.
$FV\ inflows_S$ = \$19,058.54.

Inputs	5		-10000	0	19058.54
	N	I/YR	PV	PMT	FV
Output		= 13.77			

$MIRR_S$ = 13.77%.

Project L

Inputs	5	12	0	7400	
	N	I/YR	PV	PMT	FV
Output	=				-47,011.07

$PV\ costs_L$ = \$25,000.

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FV inflows$_L$ = $47,011.07.

Inputs	5		-25000	0	47011.07
	N	I/YR	PV	PMT	FV
Output		= 13.46			

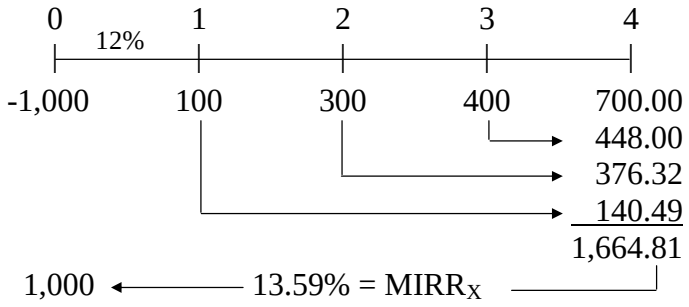
$MIRR_L$ = 13.46%.

$PI_S = \frac{\$10{,}814.33}{\$10{,}000} = 1.081$. $PI_L = \frac{\$26{,}675.34}{\$25{,}000} = 1.067$.

Thus, $NPV_L > NPV_S$, $IRR_S > IRR_L$, $MIRR_S > MIRR_L$, and $PI_S > PI_L$. The scale difference between Projects S and L results in the IRR, MIRR, and PI favoring S over L. However, NPV favors Project L; thus, L adds more value to the firm so L should be chosen.

10-11 Because both projects are the same size you can just calculate each project's MIRR and choose the project with the higher MIRR. (Remember, MIRR gives conflicting results from NPV when there are scale differences between the projects.)

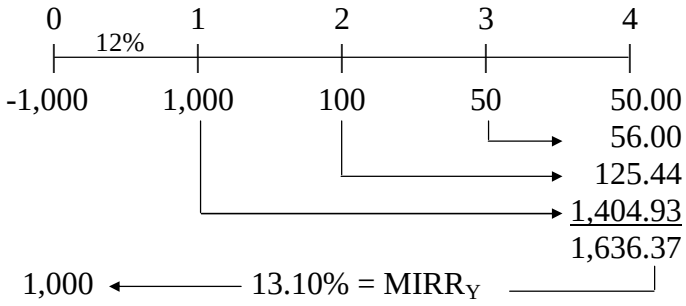
Project X:



$\$1{,}000 = \$1{,}664.81/(1 + MIRR_X)^4$.

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Project Y:



$\$1{,}000 = \$1{,}636.37/(1 + MIRR_Y)^4$.

Thus, since $MIRR_X > MIRR_Y$, Project X should be chosen.

Alternative step: You could calculate NPVs, see that Project X has the higher NPV, and just calculate $MIRR_X$.

$NPV_X = \$58.02$ and $NPV_Y = \$39.94$.

10-12 a.

Purchase price	$ 900,000
Installation	165,000
Initial outlay	$1,065,000

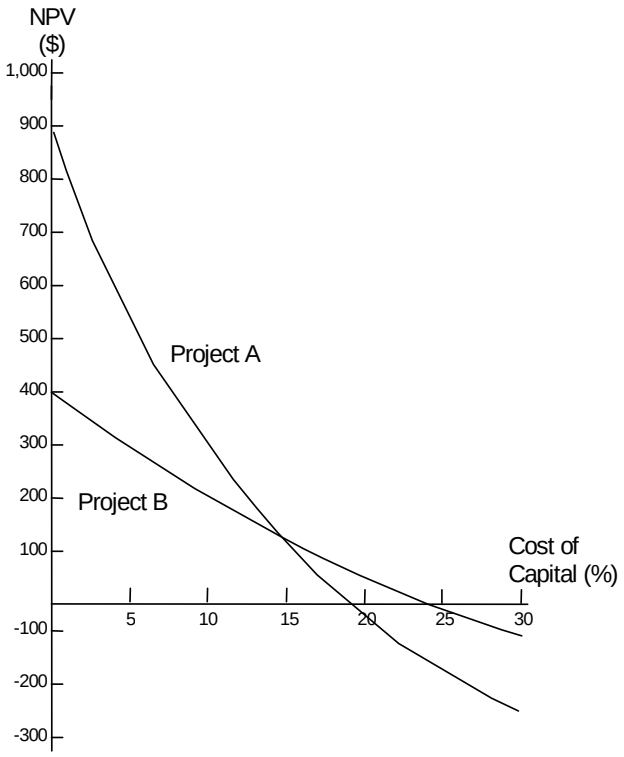
$CF_0 = -1065000$; $CF_{1-5} = 350000$; I/YR = 14; NPV = ?
NPV = $136,578; IRR = 19.22%.

b. Ignoring environmental concerns, the project should be undertaken because its NPV is positive and its IRR is greater than the firm's cost of capital.

c. Environmental effects could be added by estimating penalties or any other cash outflows that might be imposed on the firm to help return the land to its previous state (if possible). These outflows could be so large as to cause the project to have a negative NPV—in which case the project should not be undertaken.

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10-13 a.



r	NPV_A	NPV_B
0.0%	$890	$399
10.0	283	179
12.0	200	146
18.1	0	62
20.0	(49)	41
24.0	(138)	0
30.0	(238)	(51)

b. $IRR_A = 18.1\%$; $IRR_B = 24.0\%$.

c. At r = 10%, Project A has the greater NPV, specifically $283.34 as compared to Project B's NPV of $178.60. Thus, Project A would be selected. At r = 17%, Project B has an NPV of $75.95 which is higher than Project A's NPV of $31.05. Thus, choose Project B if r = 17%.

d. Here is the MIRR for Project A when r = 10%:

PV costs = $\$300 + \$387/(1.10)^1 + \$193/(1.10)^2$
$+ \$100/(1.10)^3 + \$180/(1.10)^7 = \$978.82.$

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TV inflows = $\$600(1.10)^3 + \$600(1.10)^2 + \$850(1.10)^1 = \$2{,}459.60$.

Now, MIRR is that discount rate which forces the TV of $2,459.60 in 7 years to equal $978.82:

$\$978.82 = \$2{,}459.60(1 + MIRR)^7$
$MIRR_A = 14.07\%$.

Similarly, $\$405 = \$1{,}137.28(1 + MIRR)^7$
$MIRR_B = 15.89\%$.

At r = 17%,
$MIRR_A = 17.57\%$.
$MIRR_B = 19.91\%$.

e. To find the crossover rate, construct a Project Δ which is the difference in the two projects' cash flows:

Year	Project Δ = $CF_A - CF_B$
0	$105
1	(521)
2	(327)
3	(234)
4	466
5	466
6	716
7	(180)

IRR_Δ = Crossover rate = 14.53%.

Projects A and B are mutually exclusive, thus, only one of the projects can be chosen. As long as the cost of capital is greater than the crossover rate, both the NPV and IRR methods will lead to the same project selection. However, if the cost of capital is less than the crossover rate the two methods lead to different project selections—a conflict exists. When a conflict exists the NPV method must be used.

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Because of the sign changes and the size of the cash flows, Project Δ has multiple IRRs. Thus, a calculator's IRR function will *not work*. One could use the trial and error method of entering different discount rates until NPV = $0. However, an HP can be "tricked" into giving the roots. After you have keyed Project Delta's cash flows into the CF_j register of an HP-10B, you will see an "Error-Soln" message. Now enter 10 ■ STO ■ IRR/YR and the 14.53% IRR is found. Then enter 100 ■ STO ■ IRR/YR to obtain IRR = 456.22%. Similarly, *Excel* can also be used.

10-14 a.

Year	Plan B	Plan A	Incremental Cash Flow (B – A)
0	($10,000,000)	($10,000,000)	$ 0
1	1,750,000	12,000,000	(10,250,000)
2-20	1,750,0000	0	1,750,000

If the firm goes with Plan B, it will forgo $10,250,000 in Year 1, but will receive $1,750,000 per year in Years 2-20.

b. If the firm could invest the incremental $10,250,000 at a return of 16.07%, it would receive cash flows of $1,750,000. If we set up an amortization schedule, we would find that payments of $1,750,000 per year for 19 years would amortize a loan of $10,250,000 at 16.0665%.

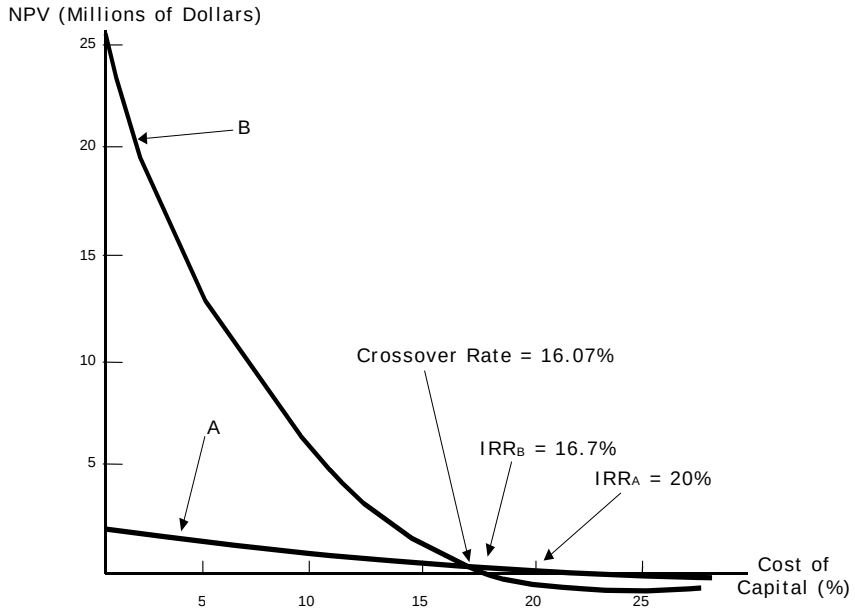
Financial calculator solution:

	N	I/YR	PV	PMT	FV
Inputs	19		-10250000	1750000	0
Output		= 16.0665			

c. Yes, assuming (1) equal risk among projects, and (2) that the cost of capital is a constant and does not vary with the amount of capital raised.

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d. See graph. If the cost of capital is less than 16.07%, then Plan B should be accepted; if r > 16.07%, then Plan A is preferred.



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10-15 a. Financial calculator solution:

Plan A

Inputs	20	10		8000000	0
	N	I/YR	PV	PMT	FV
Output			= -68,108,510		

NPV_A = \$68,108,510 – \$50,000,000 = \$18,108,510.

Plan B

Inputs	20	10		3400000	0
	N	I/YR	PV	PMT	FV
Output			= -28,946,117		

NPV_B = \$28,946,117 – \$15,000,000 = \$13,946,117.

Plan A

Inputs	20		-50000000	8000000	0
	N	I/YR	PV	PMT	FV
Output		= 15.03			

IRR_A = 15.03%.

Plan B

Inputs	20		-15000000	3400000	0
	N	I/YR	PV	PMT	FV
Output		= 22.26			

IRR_B = 22.26%.

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b. If the company takes Plan A rather than B, its cash flows will be (in millions of dollars):

Year	Cash Flows from A	Project Δ Cash Flows from B	Cash Flows
0	($50)	($15.0)	($35.0)
1	8	3.4	4.6
2	8	3.4	4.6
–	–	–	–
–	–	–	–
–	–	–	–
20	8	3.4	4.6

So, Project Δ has a "cost" of $35,000,000 and "inflows" of $4,600,000 per year for 20 years.

	N	I/YR	PV	PMT	FV
Inputs	20	10		4600000	0
Output =			-39,162,393		

NPV_{Δ} = $39,162,393 – $35,000,000 = $4,162,393.

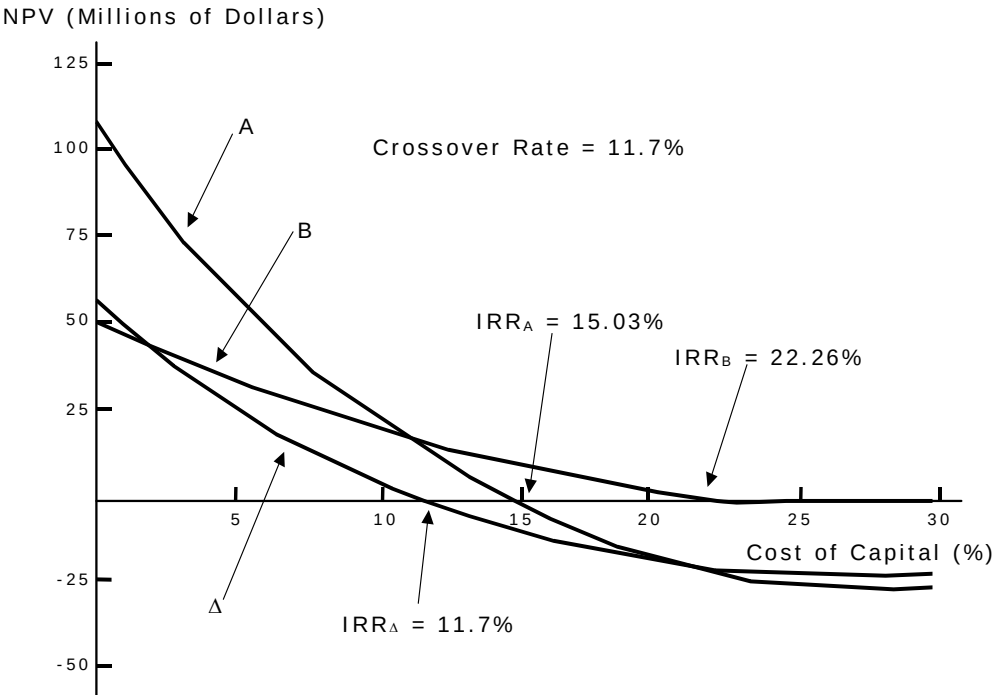
	N	I/YR	PV	PMT	FV
Inputs	20		-35000000	4600000	0
Output =		11.71			

IRR_{Δ} = 11.71%.

Since IRR_{Δ} > r, we should accept Δ. This means we should accept the larger project (Project A). In addition, when dealing with mutually exclusive projects, we use the NPV method for choosing the best project.

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c.



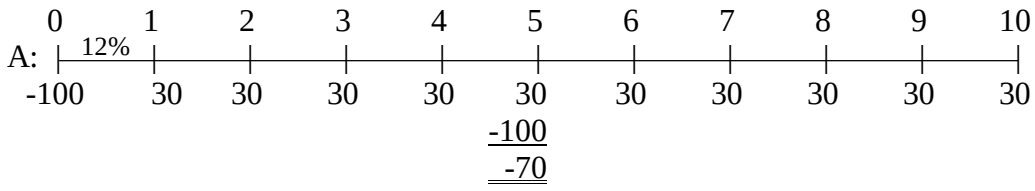
d. The NPV method implicitly assumes that the opportunity exists to reinvest the cash flows generated by a project at the cost of capital, while use of the IRR method implies the opportunity to reinvest at the IRR. If the firm's cost of capital is constant at 10%, all projects with an NPV > 0 will be accepted by the firm. As cash flows come in from these projects, the firm will either distribute them to investors, or use them as a substitute for outside capital which costs 10%. Thus, since these cash flows are expected to save the firm 10%, this is their opportunity cost reinvestment rate.

The IRR method assumes reinvestment at the internal rate of return itself, which is an incorrect assumption, given a constant expected future cost of capital, and ready access to capital markets.

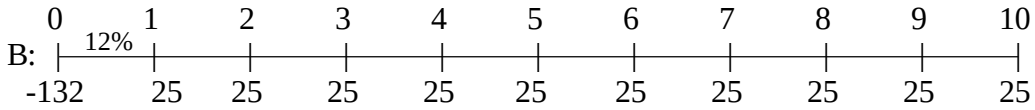
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10-16 Plane A: Expected life = 5 years; Cost = \$100 million; NCF = \$30 million; COC = 12%.

Plane B: Expected life = 10 years; Cost = \$132 million; NCF = \$25 million; COC = 12%.



Enter these values into the cash flow register: CF_0 = -100; CF_{1-4} = 30; CF_5 = -70; CF_{6-10} = 30. Then enter I/YR = 12, and press the NPV key to get NPV_A = \$12.764 million.



Enter these cash flows into the cash flow register, along with the interest rate, and press the NPV key to get NPV_B = \$9.256 million.

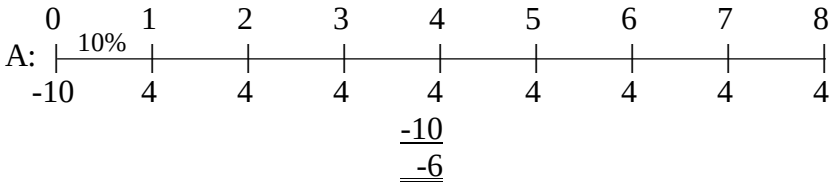
Project A is the better project and will increase the company's value by \$12.764 million.

The EAA of plane A is found by first finding the PV: N = 5, I/YR = 12, PMT = 30, FV = 0; solve for PV = \$108.143. The NPV is \$108.143 − \$100 = \$8.143 million. We convert this to an equivalent annual annuity by inputting: N = 5, I/YR = 12, PV = -8.143, FV = 0, and solve for PMT = EAA = \$2.259 million.

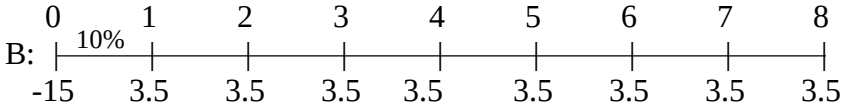
For plane B, we already found the NPV of \$9.256 million. We convert this to an equivalent annual annuity by inputting: N = 10, I/YR = 12, PV = -9.256, FV = 0, and solve for PMT = EAA = \$1.638 million.

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10-17



Machine A's simple NPV is calculated as follows: Enter CF_0 = -10 and CF_{1-4} = 4. Then enter I/YR = 10, and press the NPV key to get NPV_A = \$2.679 million. However, this does not consider the fact that the project can be repeated again. Enter these values into the cash flow register: CF_0 = -10; CF_{1-3} = 4; CF_4 = -6; CF_{5-8} = 4. Then enter I/YR = 10, and press the NPV key to get Extended NPV_A = \$4.5096 ≈ \$4.51 million.



For Machine B's NPV, enter these cash flows into the cash flow register, along with the interest rate, and press the NPV key to get NPV_B = \$3.672 ≈ \$3.67 million.

Machine A is the better project and will increase the company's value by \$4.51 million.

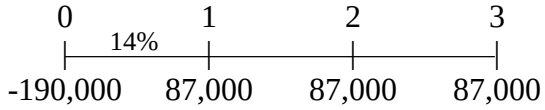
The EAA of Machine A is found by first finding the PV: N = 4, I/YR = 10, PMT = 4, FV = 0; solve for PV = \$12.679. The NPV is \$12.679 − \$10 = \$2.679 million. We convert this to an equivalent annual annuity by inputting: N = 4, I/YR = 10, PV = -2.679, FV = 0, and solve for PMT = EAA_A = 0.845 ≈ \$0.85 million.

For Machine B, we already found the NPV of \$3.672 million. We convert this to an equivalent annual annuity by inputting: N = 8, I/YR = 10, PV = -3.672, FV = 0, and solve for PMT = EAA_B = 0.688 ≈ \$0.69 million.

Again, the EAA method demonstrates that Machine A is the better project since $EAA_A > EAA_B$.

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10-18 Cash flow time line for Machine 190-3:



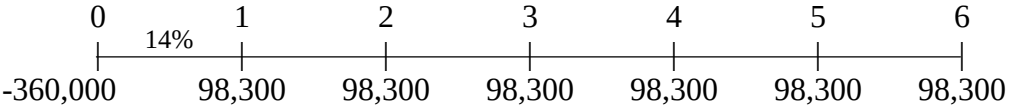
Using a financial calculator, input the following data: $CF_0 = -190000$; $CF_{1-3} = 87000$; I/YR = 14; and solve for $NPV_{190-3} = \$11{,}982$ (for 3 years).

Extended $NPV_{190-3} = \$11{,}982 + \$11{,}982/(1.14)^3 = \$20{,}070$.

EAA_{190-3}: Using a financial calculator, input the following data:

N = 3; I/YR = 14; PV = -11982; and FV = 0. Solve for PMT = EAA = \$5,161.

Cash flow time line for Machine 360-6:



Using a financial calculator, input the following data: $CF_0 = -360000$; $CF_{1-6} = 98300$; I/YR = 14; and solve for $NPV_{360-6} = \$22{,}256$ (for 6 years).

EAA_{360-6}: Using a financial calculator, input the following data:
N = 6; I/YR = 14; PV = -22256; and FV = 0. Solve for PMT = EAA = \$5,723.

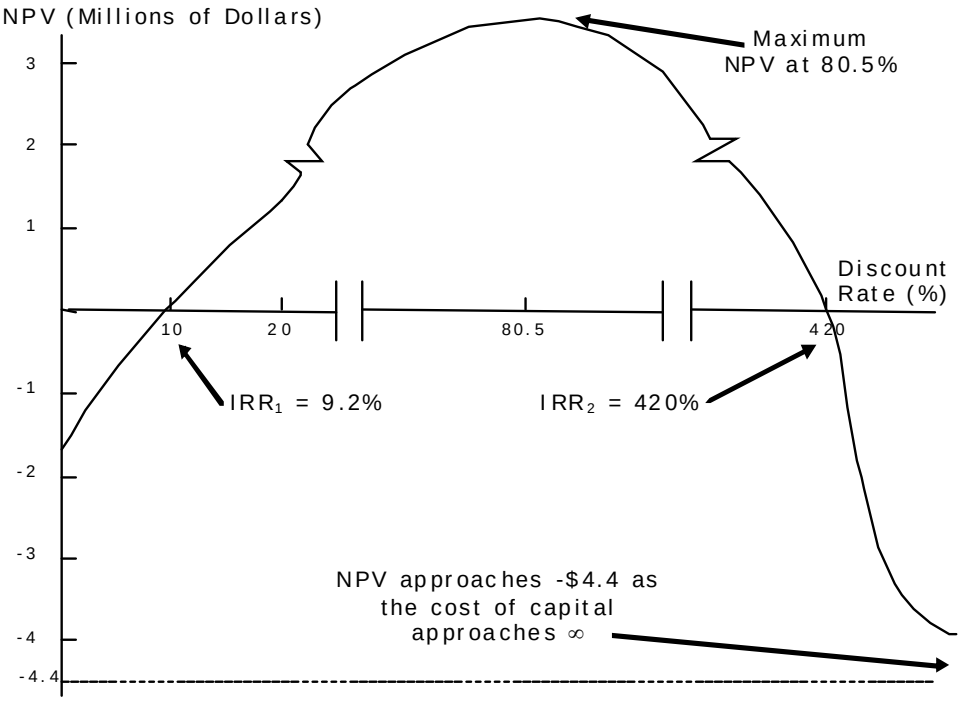
Both new machines have positive NPVs; hence, the old machine should be replaced. Further, since its NPV is greater with the replacement chain approach and its EAA is higher than Model 190-3, choose Model 360-6.

10-19 a. The project's expected cash flows are as follows (in millions of dollars):

Time	Net Cash Flow
0	($ 4.4)
1	27.7
2	(25.0)

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We can construct the following NPV profile:



Discount Rate	NPV
0%	($1,700,000)
9	(29,156)
10	120,661
50	2,955,556
100	3,200,000
200	2,055,556
300	962,500
400	140,000
410	70,204
420	2,367
430	(63,581)

The table above was constructed using a financial calculator with the following inputs: CF_0 = -4400000, CF_1 = 27700000, CF_2 = -25000000, and I/YR = discount rate to solve for the NPV.

b. If r = 8%, reject the project since NPV < 0 as shown in the project's NPV profile. But if r = 14%, accept the project because NPV > 0 as shown in the project's NPV profile.

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c. Other possible projects with multiple rates of return could be nuclear power plants where disposal of radioactive wastes is required at the end of the project's life, or leveraged leases where the borrowed funds are repaid at the end of the lease life. (See Chapter 18 of *Financial Management, 13th edition* for more information on leases.)

d. Here is the MIRR for the project when r = 8%:

PV costs = \$4,400,000 + \$25,000,000/$(1.08)^2$ = \$25,833,470.51.

TV inflows = \$27,700,000$(1.08)^1$ = \$29,916,000.00.

Now, MIRR is the discount rate that forces the PV of the TV of \$29,916,000 over 2 years to equal \$25,833,470.51:

\$25,833,470.51 = \$29,916,000($PVIF_{r,2}$).

Inputs	2		-25833470.51	0	29916000
	N	I/YR	PV	PMT	FV
Output		= 7.61			

MIRR = 7.61%.

At r = 14%, MIRR for the project is calculated as follows:

PV costs = \$4,400,000 + \$25,000,000/$(1.14)^2$ = \$23,636,688.21.

TV inflows = \$27,700,000$(1.14)^1$ = \$31,578,000.

Now, MIRR is that discount rate which forces the PV of the TV of \$31,578,000 over 2 years to equal \$23,636,688.21:

\$23,636,688.21 = \$31,578,000($PVIF_{r,2}$).

Inputs	2		-23636688.21	0	31578000
	N	I/YR	PV	PMT	FV
Output		= 15.58			

MIRR = 15.58%.

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Yes. The MIRR method leads to the same conclusion as the NPV method. Reject the project if r = 8%, which is greater than the corresponding MIRR of 7.61%, and accept the project if r = 14%, which is less than the corresponding MIRR of 15.58%.

10-20 a. The IRRs of the two alternatives are undefined. To calculate an IRR, the cash flow stream must include both cash inflows and outflows.

b. The PV of costs for the conveyor system is ($911,067), while the PV of costs for the forklift system is ($838,834). Thus, the forklift system is expected to be ($838,834) – ($911,067) = $72,233 less costly than the conveyor system, and hence the forklift trucks should be used.

Financial calculator solution:

Input: CF_0 = -500000, CF_1 = -120000, N_j = 4, CF_2 = -20000, I/YR = 8, NPV_C = ? NPV_C = -911,067.

Input: CF_0 = -200000, CF_1 = -160000, N_j = 5, I/YR = 8, NPV_F = ? NPV_F = -838,834.

10-21 a. Payback A (cash flows in thousands):

Period	Annual Cash Flows	Cumulative
0	($25,000)	($25,000)
1	5,000	(20,000)
2	10,000	(10,000)
3	15,000	5,000
4	20,000	25,000

$Payback_A$ = 2 + $10,000/$15,000 = 2.67 years.

Payback B (cash flows in thousands):

Period	Annual Cash Flows	Cumulative
0	($25,000)	($25,000)
1	20,000	(5,000)
2	10,000	5,000
3	8,000	13,000
4	6,000	19,000

$Payback_B$ = 1 + $5,000/$10,000 = 1.50 years.

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b. Discounted Payback A (cash flows in thousands):

Period	Annual Cash Flows	Discounted @10% Cash Flows	Cumulative
0	($25,000)	($25,000.00)	($25,000.00)
1	5,000	4,545.45	(20,454.55)
2	10,000	8,264.46	(12,190.09)
3	15,000	11,269.72	(920.37)
4	20,000	13,660.27	12,739.90

Discounted Payback_A = 3 + $920.37/$13,660.27 = 3.07 years.

Discounted Payback B (cash flows in thousands):

Period	Annual Cash Flows	Discounted @10% Cash Flows	Cumulative
0	($25,000)	($25,000.00)	($25,000.00)
1	20,000	18,181.82	(6,818.18)
2	10,000	8,264.46	1,446.28
3	8,000	6,010.52	7,456.80
4	6,000	4,098.08	11,554.88

Discounted Payback_B = 1 + $6,818.18/$8,264.46 = 1.825 years.

c. NPV_A = $12,739,908; IRR_A = 27.27%.
NPV_B = $11,554,880; IRR_B = 36.15%.

Both projects have positive NPVs, so both projects should be undertaken.

d. At a discount rate of 5%, NPV_A = $18,243,813.
At a discount rate of 5%, NPV_B = $14,964,829.

At a discount rate of 5%, Project A has the higher NPV; consequently, it should be accepted.

e. At a discount rate of 15%, NPV_A = $8,207,071.
At a discount rate of 15%, NPV_B = $8,643,390.

At a discount rate of 15%, Project B has the higher NPV; consequently, it should be accepted.

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f.

Year	Project Δ = $CF_A - CF_B$
0	$ 0
1	(15)
2	0
3	7
4	14

IRR_Δ = Crossover rate = 13.5254% ≈ 13.53%.

g. Use 3 steps to calculate $MIRR_A$ @ r = 10%:

Step 1: Calculate the NPV of the uneven cash inflow stream, so its FV can then be calculated. With a financial calculator, enter the cash inflow stream into the cash flow registers being sure to enter 0 for CF_0, then enter I/YR = 10, and solve for NPV = $37,739,908.

Step 2: Calculate the FV of the cash inflow stream as follows:

Enter N = 4, I/YR = 10, PV = -37739908, and PMT = 0 to solve for FV = $55,255,000.

Step 3: Calculate $MIRR_A$ as follows:
Enter N = 4, PV = -25000000, PMT = 0, and FV = 55255000 to solve for I/YR = 21.93%.

Use 3 steps to calculate $MIRR_B$ @ r = 10%:

Step 1: Calculate the NPV of the uneven cash inflow stream, so its FV can then be calculated. With a financial calculator, enter the cash inflow stream into the cash flow registers being sure to enter 0 for CF_0, then enter I/YR = 10, and solve for NPV = $36,554,880.

Step 2: Calculate the FV of the cash flow stream as follows:

Enter N = 4, I/YR = 10, PV = -36554880, and PMT = 0 to solve for FV = $53,520,000.

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Step 3: Calculate $MIRR_B$ as follows:

Enter N = 4, PV = -25000000, PMT = 0, and FV = 53520000 to solve for I/YR = 20.96%.

According to the MIRR approach, if the 2 projects were mutually exclusive, Project A would be chosen because it has the higher MIRR. This is consistent with the NPV approach.

10-22 a. NPV of termination after Year t:

$$NPV_0 = -\$22,500 + \$22,500 = 0.$$

Using a financial calculator, input the following: $CF_0 = -22500$, $CF_1 = 23750$, and I/YR = 10 to solve for $NPV_1 = -\$909.09 \approx -\$909$.

Using a financial calculator, input the following: $CF_0 = -22500$, $CF_1 = 6250$, $CF_2 = 20250$, and I/YR = 10 to solve for $NPV_2 = -\$82.64 \approx -\$83$.

Using a financial calculator, input the following: $CF_0 = -22500$, $CF_1 = 6250$, $N_j = 2$, $CF_3 = 17250$, and I/YR = 10 to solve for $NPV_3 = \$1,307.29 \approx \$1,307$.

Using a financial calculator, input the following: $CF_0 = -22500$, $CF_1 = 6250$, $N_j = 3$, $CF_4 = 11250$, and I/YR = 10 to solve for $NPV_4 = \$726.73 \approx \$727$.

Using a financial calculator, input the following: $CF_0 = -22500$, $CF_1 = 6250$, $N_j = 5$, and I/YR = 10 to solve for $NPV_5 = \$1,192.42 \approx \$1,192$.

The firm should operate the truck for 3 years, $NPV_3 = \$1,307$.

b. No. Salvage possibilities could only raise NPV and IRR. The value of the firm is maximized by terminating the project after Year 3.

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SOLUTION TO SPREADSHEET PROBLEM

10-23 The detailed solution for the problem is available in the file ***Solution for Ch10 P23 Build a Model.xls*** at the textbook's Web site.

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MINI CASE

You have just graduated from the MBA program of a large university, and one of your favorite courses was "Today's Entrepreneurs." In fact, you enjoyed it so much you have decided you want to "be your own boss." While you were in the master's program, your grandfather died and left you $1 million to do with as you please. You are not an inventor and you do not have a trade skill that you can market; however, you have decided that you would like to purchase at least one established franchise in the fast-foods area, maybe two (if profitable). The problem is that you have never been one to stay with any project for too long, so you figure that your time frame is three years. After three years you will sell off your investment and go on to something else.

You have narrowed your selection down to two choices; (1) Franchise L, Lisa's Soups, Salads, & Stuff and (2) Franchise S, Sam's Fabulous Fried Chicken. The net cash flows shown below include the price you would receive for selling the franchise in Year 3 and the forecast of how each franchise will do over the three-year period. Franchise L's cash flows will start off slowly but will increase rather quickly as people become more health conscious, while Franchise S's cash flows will start off high but will trail off as other chicken competitors enter the marketplace and as people become more health conscious and avoid fried foods. Franchise L serves breakfast and lunch, while Franchise S serves only dinner, so it is possible for you to invest in both franchises. You see these franchises as perfect complements to one another: You could attract both the lunch and dinner crowds and the health conscious and not so health conscious crowds without the franchises directly competing against one another.

Here are the net cash flows (in thousands of dollars):

	Expected Net Cash Flows	
Year	**Franchise L**	**Franchise S**
0	**($100)**	**($100)**
1	**10**	**70**
2	**60**	**50**
3	**80**	**20**

Depreciation, salvage values, net working capital requirements, and tax effects are all included in these cash flows.

You also have made subjective risk assessments of each franchise, and concluded that both franchises have risk characteristics that require a return of 10%. You must now determine whether one or both of the franchises should be accepted.

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a. What is capital budgeting?

Answer: Capital budgeting is the process of analyzing additions to fixed assets. Capital budgeting is important because, more than anything else, fixed asset investment decisions chart a company's course for the future. Conceptually, the capital budgeting process is identical to the decision process used by individuals making investment decisions. These steps are involved:

1. Estimate the cash flows—interest and maturity value or dividends in the case of bonds and stocks, operating cash flows in the case of capital projects.

2. Assess the riskiness of the cash flows.

3. Determine the appropriate discount rate, based on the riskiness of the cash flows and the general level of interest rates. This is called the project cost of capital in capital budgeting.

4. Evaluate the cash flows.

b. What is the difference between independent and mutually exclusive projects?

Answer: Projects are independent if the cash flows of one are not affected by the acceptance of the other. Conversely, two projects are mutually exclusive if acceptance of one impacts adversely the cash flows of the other; that is, at most one of two or more such projects may be accepted. Put another way, when projects are mutually exclusive it means that they do the same job. For example, a forklift truck versus a conveyor system to move materials, or a bridge versus a ferry boat.

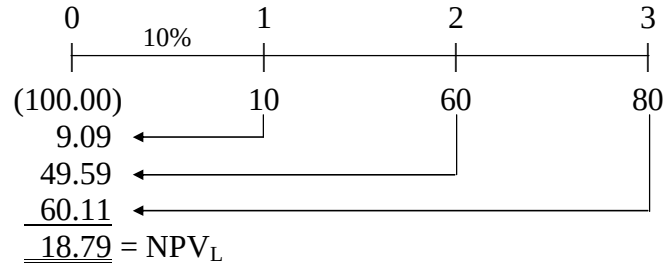
c. 1. Define the term *net present value (NPV)*. What is each franchise's NPV?

Answer: The net present value (NPV) is simply the sum of the present values of a project's cash flows:

$$NPV = \sum_{t=0}^{N} \frac{CF_t}{(1+r)^t}.$$

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Franchise L's NPV is $18.79:



NPVs are easy to determine using a calculator with an NPV function. Enter the cash flows sequentially, with outflows entered as negatives; enter the cost of capital; and then press the NPV button to obtain the franchise's NPV, $18.78 (note the penny rounding difference). The NPV of Franchise S is NPV_S = $19.98.

c. 2. What is the rationale behind the NPV method? According to NPV, which franchise or franchises should be accepted if they are independent? Mutually exclusive?

Answer: The rationale behind the NPV method is straightforward: if a project has NPV = $0, then the project generates exactly enough cash flows (1) to recover the cost of the investment and (2) to enable investors to earn their required rates of return (the opportunity cost of capital). If NPV = $0, then in a financial (but not an accounting) sense, the project breaks even. If the NPV is positive, then more than enough cash flow is generated, and conversely if NPV is negative.

Consider Franchise L's cash inflows, which total $150. They are sufficient (1) to return the $100 initial investment, (2) to provide investors with their 10% aggregate opportunity cost of capital, and (3) to still have $18.79 left over on a present value basis. This $18.79 excess PV belongs to the shareholders—the debtholders' claims are fixed, so the shareholders' wealth will be increased by $18.79 if Franchise L is accepted. Similarly, shareholders gain $19.98 in value if Franchise S is accepted.

If Franchises L and S are independent, then both should be accepted, because they both add to shareholders' wealth, hence to the stock price. If the franchises are mutually exclusive, then Franchise S should be chosen over L, because S adds more to the value of the firm.

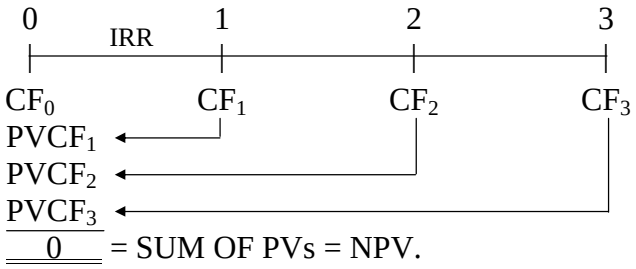
c. 3. Would the NPVs change if the cost of capital changed?

Answer: The NPV of a project is dependent on the cost of capital used. Thus, if the cost of capital changed, the NPV of each franchise would change. NPV declines as r increases, and NPV rises as r decreases.

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d. 1. Define the term *internal rate of return (IRR)*. What is each franchise's IRR?

Answer: The internal rate of return (IRR) is the discount rate that forces the NPV of a project to equal zero:

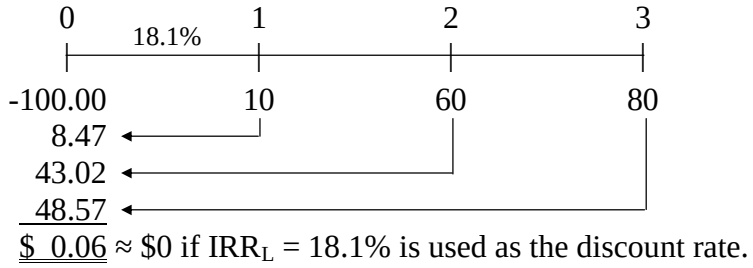


Expressed as an equation, we have:

$$\text{IRR: } \sum_{t=0}^{N} \frac{CF_t}{(1+\text{IRR})^t} = \$0 = \text{NPV}.$$

Note that the IRR equation is the same as the NPV equation, except that to find the IRR the equation is solved for the particular discount rate, IRR, which forces the project's NPV to equal zero (the IRR) rather than using the cost of capital (r) in the denominator and finding NPV. Thus, the two approaches differ in only one respect: in the NPV method, a discount rate is specified (the project's cost of capital) and the equation is solved for NPV, while in the IRR method, the NPV is specified to equal zero and the discount rate (IRR) that forces this equality is found.

Franchise L's IRR is 18.1%:



Therefore, $IRR_L \approx 18.1\%$.

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A financial calculator is extremely helpful when calculating IRRs. The cash flows are entered sequentially, and then the IRR button is pressed. For Franchise S, $IRR_S \approx$ 23.6%. Note that with many calculators, you can enter the cash flows into the cash flow register, also enter r = I/YR, and then calculate both NPV and IRR by pressing the appropriate buttons.

d. 2. How is the IRR on a project related to the YTM on a bond?

Answer: The IRR is to a capital project what the YTM is to a bond. It is the expected rate of return on the project, just as the YTM is the promised rate of return on a bond.

d. 3. What is the logic behind the IRR method? According to IRR, which franchises should be accepted if they are independent? Mutually exclusive?

Answer: IRR measures a project's profitability in the rate of return sense: If a project's IRR equals its cost of capital, then its cash flows are just sufficient to provide investors with their required rates of return. An IRR greater than r implies an economic profit, which accrues to the firm's shareholders, while an IRR less than r indicates an economic loss, or a project that will not earn enough to cover its cost of capital.

Projects' IRRs are compared to their costs of capital, or hurdle rates. Since Franchises L and S both have a hurdle rate of 10%, and since both have IRRs greater than that hurdle rate, both should be accepted if they are independent. However, if they are mutually exclusive, Franchise S would be selected, because it has the higher IRR.

d. 4. Would the franchises' IRRs change if the cost of capital changed?

Answer: IRRs are independent of the cost of capital. Therefore, neither IRR_S nor IRR_L would change if r changed. However, the acceptability of the franchises could change—L would be rejected if r were above 18.1%, and S would also be rejected if r were above 23.6%.

e. 1. Draw NPV profiles for Franchises L and S. At what discount rate do the profiles cross?

Answer: The NPV profiles are plotted in the figure below.

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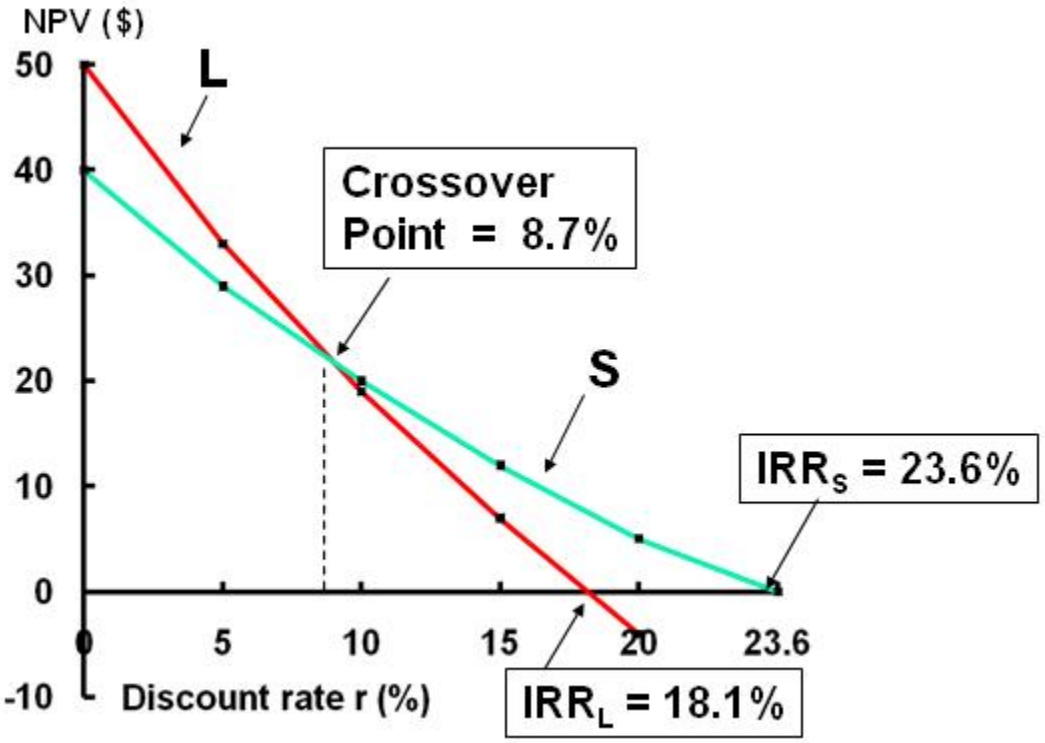
Note the following points:

1. The Y-intercept is the project's NPV when r = 0%. This is $50 for L and $40 for S.

2. The X-intercept is the project's IRR. This is 18.1% for L and 23.6% for S.

3. NPV profiles are curves rather than straight lines. To see this, note that these profiles approach cost = -$100 as r approaches infinity.

4. From the figure below, it appears that the crossover rate is between 8% and 9%. The precise value is approximately 8.7%. One can calculate the crossover rate by (1) going back to the data on the problem, finding the cash flow differences for each year, (2) entering those differences into the cash flow register, and (3) pressing the IRR button to get the crossover rate, 8.68% ≈ 8.7%.



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r	NPV_L	NPV_S
0%	$50	$40
5	33	29
10	19	20
15	7	12
20	(4)	5

e. 2. Look at your NPV profile graph without referring to the actual NPVs and IRRs. Which franchise or franchises should be accepted if they are independent? Mutually exclusive? Explain. Are your answers correct at any cost of capital less than 23.6%?

Answer: The NPV profiles show that the IRR and NPV criteria lead to the same accept/reject decision for any independent project. Consider Franchise L. It intersects the X-axis at its IRR, 18.1%. According to the IRR rule, L is acceptable if r is less than 18.1%. Also, at any r less than 18.1%, L's NPV profile will be above the X-axis, so its NPV will be greater than $0. Thus, for any independent project, NPV and IRR lead to the same accept/reject decision.

Now assume that L and S are mutually exclusive. In this case, a conflict might arise. First, note that $IRR_S = 23.6\% > 18.1\% = IRR_L$. Therefore, regardless of the size of r, Franchise S would be ranked higher by the IRR criterion. However, the NPV profiles show that $NPV_L > NPV_S$ if r is less than 8.7%. Therefore, for any r below the 8.7% crossover rate, say r = 7%, the NPV rule says choose L, but the IRR rule says choose S. Thus, if r is less than the crossover rate, a ranking conflict occurs.

f. 1. What is the underlying cause of ranking conflicts between NPV and IRR?

Answer: For normal projects' NPV profiles to cross, one project must have both a higher vertical axis intercept and a steeper slope than the other. A project's vertical axis intercept typically depends on (1) the size of the project and (2) the size and timing pattern of the cash flows—large projects, and ones with large distant cash flows, would generally be expected to have relatively high vertical axis intercepts. The slope of the NPV profile depends entirely on the timing pattern of the cash flows—long-term projects have steeper NPV profiles than short-term ones. Thus, we conclude that NPV profiles can cross in two situations: (1) when mutually exclusive projects differ in scale (or size) and (2) when the projects' cash flows differ in terms of the timing pattern of their cash flows (as for Franchises L and S).

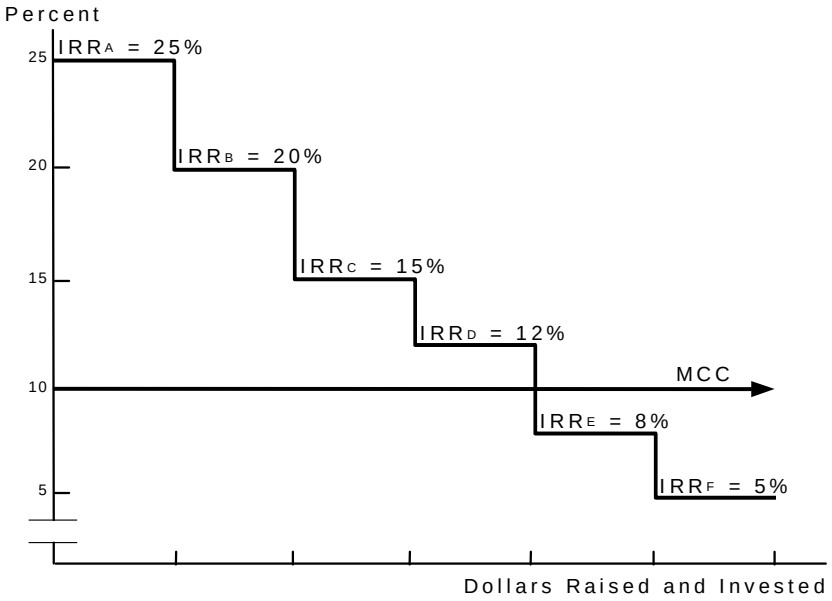
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f. 2. What is the "reinvestment rate assumption," and how does it affect the NPV versus IRR conflict?

Answer: The underlying cause of ranking conflicts is the reinvestment rate assumption. All DCF methods implicitly assume that cash flows can be reinvested at some rate, regardless of what is actually done with the cash flows. Discounting is the reverse of compounding. Since compounding assumes reinvestment, so does discounting. NPV and IRR are both found by discounting, so they both implicitly assume some discount rate. Inherent in the NPV calculation is the assumption that cash flows can be reinvested at the project's cost of capital, while the IRR calculation assumes reinvestment at the IRR rate.

f. 3. Which method is the best? Why?

Answer: Whether NPV or IRR gives better rankings depends on which has the better reinvestment rate assumption. Normally, the NPV's assumption is better. The reason is as follows: A project's cash inflows are generally used as substitutes for outside capital, that is, projects' cash flows replace outside capital and, hence, save the firm the cost of outside capital. Therefore, in an opportunity cost sense, a project's cash flows are reinvested at the cost of capital. To see this graphically, think of the following situation: Assume the firm's cost of capital is a constant 10% within the relevant range of financing considered, and it has projects available as shown in the graph below:



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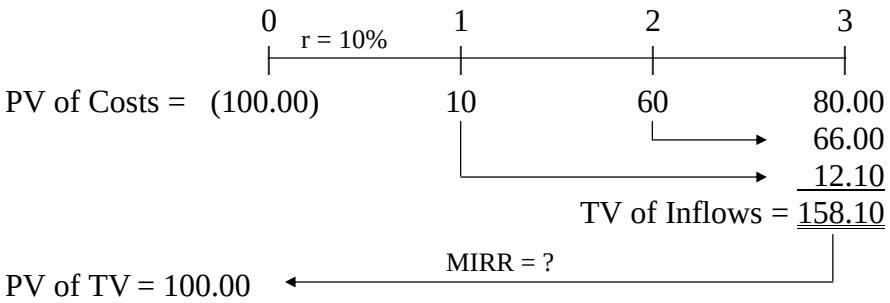
What projects will be accepted, by either NPV or IRR? Projects A, B, C, and D.

If the same situation exists year after year, at what rate of return will cash flows from earlier years' investments be reinvested? Capital budgeting decisions are made in this sequence: (1) The company would say, "We can take on A, B, C, and D and finance them with 10% money, so let's do it." (2) Then, it would get cash flows from earlier years' projects. What would it do with those cash flows? It would use them in lieu of raising money that costs 10%, so it would save 10%. Therefore, 10% is the opportunity cost of the cash flows. In effect, cash flows are reinvested at the 10% cost of capital.

Note, however, that NPV and IRR always give the same accept/reject decisions for independent projects, so IRR can be used just as well as NPV when independent projects are being evaluated. The NPV versus IRR conflict arises only if mutually exclusive projects are involved.

g. 1. Define the term *modified IRR (MIRR)*. Find the MIRRs for Franchises L and S.

Answer: MIRR is the discount rate that equates the present value of the terminal value of the inflows, compounded at the cost of capital, to the present value of the costs. Here is the setup for calculating Franchise L's modified IRR:



$$\$100 = \frac{\$158.10}{(1+\text{MIRR})^3}.$$

$$\text{PV costs} = \frac{\text{TV}}{(1+\text{MIRR})^N} = \sum_{t=0}^{N} \frac{\text{COF}_t}{(1+r)^t} = \frac{\sum_{t=1}^{N} \text{CIF}_t (1+r)^{N-t}}{(1+\text{MIRR})^N}.$$

After you calculate the TV, enter N = 3, PV = -100, PMT = 0, FV = 158.1, and then press I/YR to get the answer, $MIRR_L = 16.5\%$. We could calculate $MIRR_S$ similarly: $MIRR_S = 16.9\%$. Thus, Franchise S is ranked higher than L. This result is consistent with the NPV decision.

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g. 2. What are the MIRR's advantages and disadvantages vis-a-vis the regular IRR? What are the MIRR's advantages and disadvantages vis-a-vis the NPV?

Answer: MIRR is a better rate of return measure than IRR for two reasons: (1) It correctly assumes reinvestment at the project's cost of capital rather than at its IRR. (2) MIRR avoids the problem of multiple IRRs—there can be only one MIRR for a given project.

MIRR does not always lead to the same decision as NPV when mutually exclusive projects are being considered. In particular, small projects often have a higher MIRR, but a lower NPV, than larger projects. Thus, MIRR is not a perfect substitute for NPV, and NPV remains the single best decision rule. However, MIRR is superior to the regular IRR, and if a rate of return measure is needed, MIRR should be used.

Business executives agree. As noted in the text, business executives prefer to compare projects' rates of return to comparing their NPVs. This is an empirical fact. As a result, financial managers are substituting MIRR for IRR in their discussions with other corporate executives. This fact was brought out in the October 1989 FMA meetings, where executives from DuPont, Hershey, and Ameritech, among others, all reported a switch from IRR to MIRR.

h. What does the profitability index (PI) measure? What are the PI's for Franchises S and L?

Answer: The PI is equal to the present value of all future cash flows divided by the initial cost. It measures the "bang for the buck."

PI_S = \$119.98 / \$100 = 1.1998.

PI_L = \$118.78 / \$100 = 1.1878.

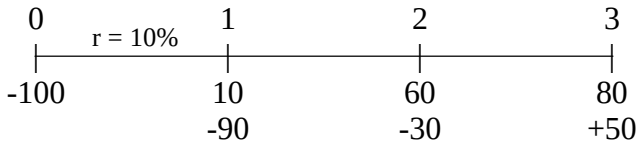
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i. 1. What is the payback period? Find the paybacks for Franchises L and S.

Answer: The payback period is the expected number of years required to recover a project's cost. We calculate the payback by developing the cumulative cash flows as shown below for Franchise L (in thousands of dollars):

	Expected NCF	
Year	Annual	Cumulative
0	($100)	($100)
1	10	(90)
2	60	(30)
3	80	50

Payback is between t = 2 and t = 3



Franchise L's $100 investment has not been recovered at the end of Year 2, but it has been more than recovered by the end of Year 3. Thus, the recovery period is between 2 and 3 years. If we assume that the cash flows occur evenly over the year, then the investment is recovered $30/$80 = 0.375 ≈ 0.4 into Year 3. Therefore, $Payback_L$ = 2.4 years. Similarly, $Payback_S$ = 1.6 years.

i. 2. What is the rationale for the payback method? According to the payback criterion, which franchise or franchises should be accepted if the firm's maximum acceptable payback is 2 years, and if Franchises L and S are independent? If they are mutually exclusive?

Answer: Payback represents a type of "breakeven" analysis: The payback period tells us when the project will break even in a cash flow sense. With a required payback of 2 years, Franchise S is acceptable, but Franchise L is not. Whether the two projects are independent or mutually exclusive makes no difference in this case.

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i. 3. What is the difference between the regular and discounted payback periods?

Answer: Discounted payback is similar to payback except that discounted cash flows are used.

Setup for Franchise L's discounted payback, assuming a 10% cost of capital:

	Expected Net Cash Flows		
Year	Cash Flows	Discounted Cash Flows	Cumulative Cash Flows
0	($100)	($100.00)	($100.00)
1	10	9.09	(90.91)
2	60	49.59	(41.32)
3	80	60.11	18.79

Discounted $\text{Payback}_L = 2 + (\$41.32/\$60.11) = 2.69 = 2.7$ years.

Versus 2.4 years for the regular payback.

i. 4. What is the main disadvantage of discounted payback? Is the payback method of any real usefulness in capital budgeting decisions?

Answer: Regular payback has 3 critical deficiencies: (1) It ignores the time value of money, (2) it ignores the cash flows that occur after the payback period, and (3) it does not provide a specific acceptance rule. Discounted payback does consider the time value of money, but it still fails to consider cash flows after the payback period and it does not provide a specific acceptance rule; so it still has basic flaws. In spite of these deficiencies, many firms today still calculate the discounted payback and give some weight to it when making capital budgeting decisions. However, payback is not generally used as the primary decision tool. Rather, it is used as a rough measure of a project's liquidity and riskiness.

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j. **As a separate project (Project P), you are considering sponsoring a pavilion at the upcoming World's Fair. The pavilion would cost $800,000, and it is expected to result in $5 million of incremental cash inflows during its 1 year of operation. However, it would then take another year, and $5 million of costs, to demolish the site and return it to its original condition. Thus, Project P's expected net cash flows look like this (in millions of dollars):**

Year	Net Cash Flows
0	($0.8)
1	5.0
2	(5.0)

The project is estimated to be of average risk, so its cost of capital is 10%.

j. 1. What are normal and nonnormal cash flows?

Answer: Normal cash flows begin with a negative cash flow (or a series of negative cash flows), switch to positive cash flows, and then remain positive. They have only one change in sign. (Note: normal cash flows can also start with positive cash flows, switch to negative cash flows, and then remain negative.) Nonnormal cash flows have more than one sign change. For example, they may start with negative cash flows, switch to positive, and then switch back to negative.

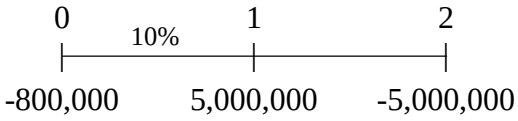
Projects with normal cash flows have outflows, or costs, in the first year (or years) followed by a series of inflows. Projects with nonnormal cash flows have one or more outflows after the inflow stream has begun. Here are some examples:

	Inflow (+) Or Outflow (–) In Year					
	0	1	2	3	4	5
Normal	–	+	+	+	+	+
	–	–	+	+	+	+
	–	–	–	+	+	+
Nonnormal	–	+	+	+	+	–
	–	+	+	–	+	–
	+	+	+	–	–	–

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j. 2. What is Project P's NPV? What is its IRR? Its MIRR?

Answer: Here is the time line for the cash flows, and the NPV:



NPV_P = -$386,776.86.

We can find the NPV by entering the cash flows into the cash flow register, entering I/YR = 10, and then pressing the NPV button. However, calculating the IRR presents a problem. With the cash flows in the register, press the IRR button. An HP-10B financial calculator will give the message "error-soln." This means that Project P has multiple IRRs. An HP-17B will ask for a guess. If you guess 10%, the calculator will produce IRR = 25%. If you guess a high number, such as 200%, it will produce the second IRR, 400%[1]. The MIRR of Project P = 5.6%, and is found by computing the discount rate that equates the terminal value ($5.5 million) to the present value of cost ($4.93 million).

j. 3. Draw Project P's NPV profile. Does Project P have normal or nonnormal cash flows? Should this project be accepted?

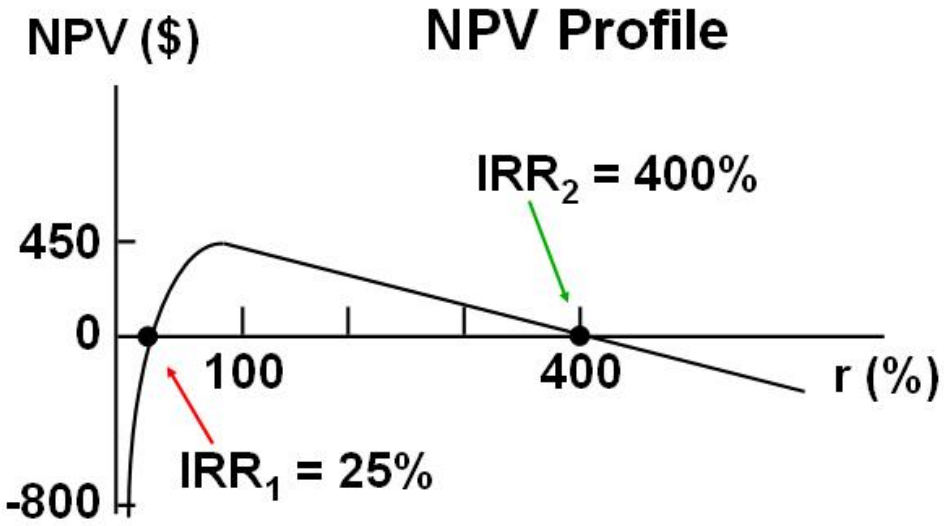
Answer: You could put the cash flows in your calculator and then enter a series of r values, get an NPV for each, and then plot the points to construct the NPV profile. We used a spreadsheet program to automate the process and then to draw the profile. Note that the profile crosses the X-axis twice, at 25% and at 400%, signifying two IRRs. Which IRR is correct? In one sense, they both are—both cause the project's NPV to equal zero. However, in another sense, both are wrong—neither has any economic or financial significance.

Project P has nonnormal cash flows; that is, it has more than one change of signs in the cash flows. Without this nonnormal cash flow pattern, we would not have the multiple IRRs.

Since Project P's NPV is negative, the project should be rejected, even though both IRRs (25% and 400%) are greater than the project's 10% cost of capital. The MIRR of 5.6% also supports the decision that the project should be rejected.

[1] Looking at the figure below, if you guess an IRR to the left of the peak NPV rate, the lower IRR will appear. If you guess IRR > peak NPV rate, the higher IRR will appear.

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k. **In an unrelated analysis, you have the opportunity to choose between the following two mutually exclusive projects:**

	Expected Net Cash Flows	
Year	**Project S**	**Project L**
0	**($100,000)**	**($100,000)**
1	**60,000**	**33,500**
2	**60,000**	**33,500**
3	**—**	**33,500**
4	**—**	**33,500**

The projects provide a necessary service, so whichever one is selected is expected to be repeated into the foreseeable future. Both projects have a 10% cost of capital.

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k. 1. What is each project's initial NPV without replication?

Answer: The NPVs, found with a financial calculator, are calculated as follows:

Input the following: $CF_0 = -100000$, $CF_1 = 60000$, $N_J = 2$, and I/YR = 10 to solve for $NPV_S = \$4{,}132.23 \approx \$4{,}132$.

Input the following: $CF_0 = -100000$, $CF_1 = 33500$, $N_J = 4$, and I/YR = 10 to solve for $NPV_L = \$6{,}190.49 \approx \$6{,}190$.

However, if we make our decision based on the initial NPVs, we would be biasing the decision against the shorter project. Since the projects are expected to be replicated, if we initially choose Project S, it would be repeated after 2 years. However, the initial NPVs do not reflect the replication cash flows.

k. 2. What is each project's equivalent annual annuity?

Answer: We begin with the NPVs found in the previous step. We then find the annuity payment stream that has the same present value as follows:

For Project S, input the following: N = 2, I/YR = 10, PV = −4,132.23, FV = 0, and solve for PMT = EAA = \$2,380.95.

For Project L, input the following: N = 4, I/YR = 10, PV = −6,190.49, FV = 0, and solve for PMT = EAA = \$1,952.92.

Project S is preferred because it has a higher EAA.

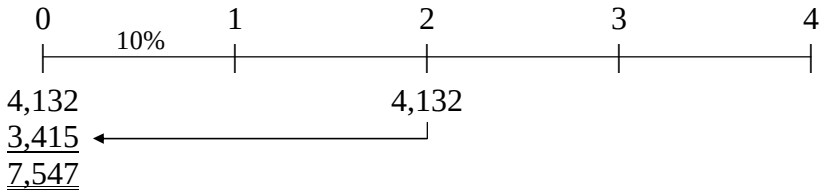
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k. 3. Now apply the replacement chain approach to determine the projects' extended NPVs. Which project should be chosen?

Answer: The simple replacement chain approach assumes that the projects will be replicated out to a common life. Since Project S has a 2-year life and L has a 4-year life, the shortest common life is 4 years. Project L's common-life NPV is its initial NPV:

$$\text{Common-Life NPV}_L = \$6{,}190.$$

However, Project S would be replicated in Year 2, and if we assume that the replicated project's cash flows are identical to the first set of cash flows, then the replicated NPV is also $4,132, but it arrives in Year 2. We can put Project S's cash flow situation on a time line:



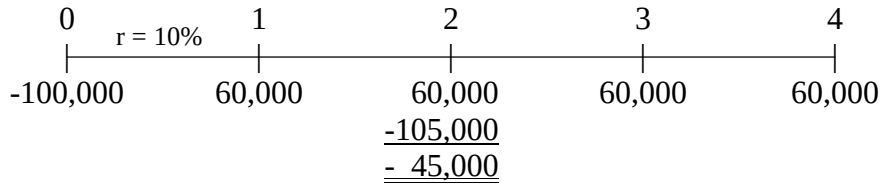
Here we see that S's common-life NPV is $NPV_S = \$7{,}547$.

Thus, when compared over a 4-year common life, Project S has the higher NPV, hence it should be chosen. Project S would have the higher NPV over any common life.

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k. 4. Now assume that the cost to replicate Project S in 2 years will increase to $105,000 because of inflationary pressures. How should the analysis be handled now, and which project should be chosen?

Answer: If the cost of Project S is expected to increase, the replication project is not identical to the original, and the EAA approach cannot be used. In this situation, we would put the cash flows on a time line as follows:



Common-Life NPV_S = $3,415.

With this change, the common-life NPV of Project S is less than that for Project L, and hence Project L should be chosen.

l. You are also considering another project which has a physical life of 3 years; that is, the machinery will be totally worn out after 3 years. However, if the project were terminated prior to the end of 3 years, the machinery would have a positive salvage value. Here are the project's estimated cash flows:

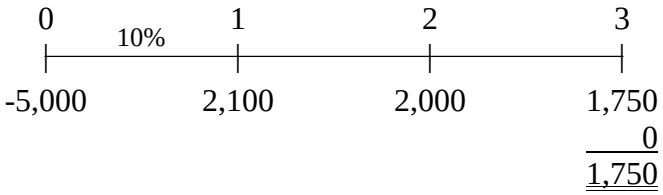
Year	Initial Investment And Operating Cash Flows	End-of-Year Net Salvage Value
0	($5,000)	$5,000
1	2,100	3,100
2	2,000	2,000
3	1,750	0

Using the 10% cost of capital, what is the project's NPV if it is operated for the full 3 years? Would the NPV change if the company planned to terminate the project at the end of Year 2? At the end of Year 1? What is the project's optimal (economic) life?

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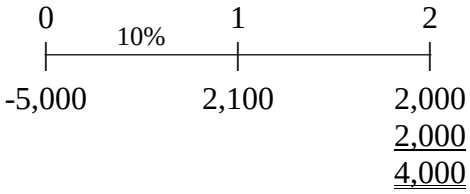
Answer: Here are the time lines for the 3 alternative lives:

No termination:



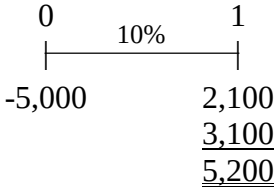
NPV = -$123.

Terminate after 2 years:



NPV = $215.

Terminate after 1 year:



NPV = -$273.

We see (1) that the project is acceptable only if operated for 2 years, and (2) that a project's engineering life does not always equal its economic life.

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m. **After examining all the potential projects, you discover that there are many more projects this year with positive NPVs than in a normal year. What two problems might this extra large capital budget cause?**

You only have a limited amount of capital to commit to projects. If you have to raise external capital to fund some of these other positive NPV projects, then you may be faced with an increasing cost of capital. This is called an increasing marginal cost of capital schedule, and it also happens to companies when they exhaust their internal sources of funds and have to go to external capital markets for their funding. This increased cost of capital may cause you to reject projects that you might otherwise accept because with your increased cost of capital, some projects may now have a negative NPV when they would otherwise have a positive NPV in a normal year.

Another effect of this large capital budget is that you may choose to ration capital—i.e. not fund all of the projects. This is called capital rationing, and companies and investors do this when for whatever reason they put a cap on the funds they are willing to invest in new projects.

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